

Project Coversheet

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Project Title (Example – Week1, Week2, Week3, Week 4)	Week1

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

Week 1: Project: Data Analysis for Business Insights

Project Title: Customer Sign-Up Behaviour & Data Quality Audit

Introduction

This report is made as part of a data quality audit and behavioural analysis for Rapid Scale, a fast-growing SaaS company offering tiered subscription plans. As a member of the Business Intelligence team, the objective of this analysis is to support the Monthly Business Review (MBR) by evaluating customer sign-up data. The dataset contains information on user acquisition sources, subscription plans, demographics, sign-up dates and marketing opt-in behaviours. The analysis focuses on identifying data quality issues, understanding user acquisition trends and assessing customer engagement patterns. The insights produced aim to support the Marketing and Onboarding teams in optimising campaigns and improving user engagement workflows.

Data cleaning summary

The first steps included understanding the dataset structure, data types and extent of missing values. A duplicate record was identified using the customer ID and removed to prevent double counting of users. Several attributes had missing data, including age, region, email and acquisition source. Missing categorical values were grouped under an “Unknown” category to ensure affected records were retained for analysis. Text-based (object) attributes such as plan_selected, gender, region and marketing opt-in status were standardised to ensure consistent formatting and naming conventions, including correcting inconsistent labels. Date was converted to datetime data type format and age values were converted to numeric format to support analysis. These cleaning steps improved data consistency and reliability while preserving overall integrity of the dataset.

```
#handle missing values
df.isna().sum() #check for missing values

# Age values are left as null to avoid introducing bias
# Email missing values retained as they do not affect behavioural analysis
# Region missing values were previously handled using an 'Unknown' category
```

	0
customer_id	1
name	9
email	34
signup_date	6
source	9
region	0
plan_selected	0
marketing_opt_in	0
age	12
gender	0

dtype: int64

Figure 1: Missing values by column after data cleaning

Key findings & trends

The analysis of customer sign-up data shows that YouTube is the strongest acquisition channel, as there are more user sign-ups compared to the other sources. This suggests, video-based marketing is more effective to attract new users.

Weekly sign-up activity remains relatively stable over time, with similar volumes observed across most weeks, there are no ups and down in numbers, this indicates steadiness and sustained user acquisition rather than reliance on short-term campaigns.

In terms of subscription choices, higher-tier plans are more popular, with the “Premium” plan selected more frequently, followed by “Pro” plan. This suggests that users see value in advanced features, presenting an opportunity to continue promoting premium features.

```
#sign-ups by source
df['source'].value_counts() #counts sign-ups per source
```

	count
source	
Youtube	58
Google	50
Referral	49
Instagram	48
Facebook	40
Linkedin	39
??	6

dtype: int64

Figure 2: Number of customer sign-ups by acquisition source

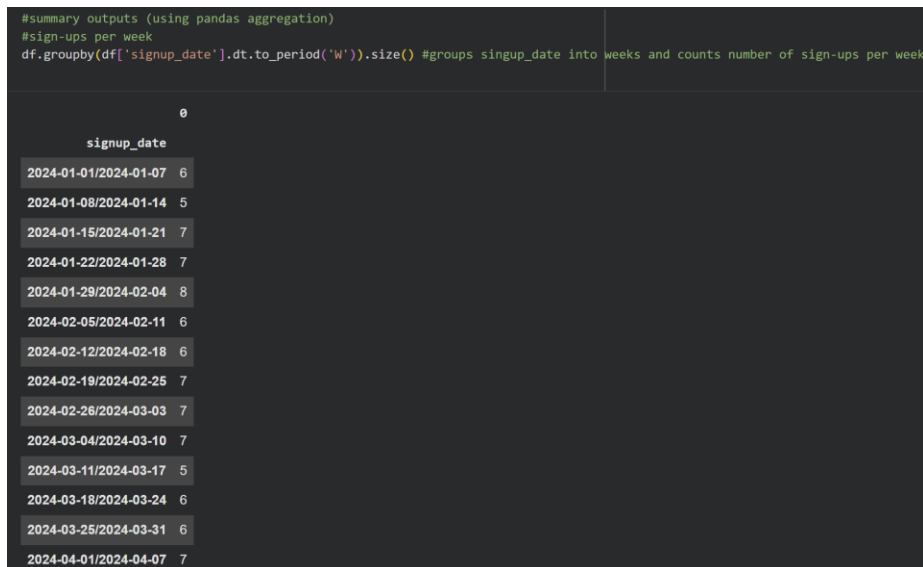


Figure 3: Weekly customer sign-ups

Business question answers

1. Which acquisition source brought in the most users last month?

Based on the sign-up data, YouTube is the acquisition source that brought-in the highest number of users. When reviewing overall sign-ups and recent weekly trends in the dataset, YouTube consistently records more user registrations than other sources such as “Google”, “Instagram” and “Referral”. Although the analysis focused on weekly and overall summaries rather than showing calendar months, the results indicate YouTube contributed to the increase in user sign-ups. A small number of records contain an “unknown” source category, which limits the attribution accuracy; however, this does not affect the overall conclusion.

2. Which region shows signs of missing or incomplete data?

The region attribute shows signs of missing or incomplete data. During the data cleaning process, several records were found to be missing region values and were grouped under an “Unknown” category. This category suggests gaps in data collection at sign-up stage rather than representing a true geographic region. As a result, analysis of regional performance and customer distribution is limited. Improving how regional data is collected in future would allow for more accurate geographic insights and reporting.

3. Are older users more or less likely to opt in to marketing?

Based on the available data, there is no clear evidence that older users are more or less likely to opt-in to marketing communications. Marketing opt-in responses appear generally balanced across the dataset. However, the age variable contains missing values and at least one unrealistic entry, which reduces the reliability of age-based comparisons. As a result, it is not possible to draw a concrete conclusion about the relationship between age and marketing opt-in behaviour. More complete and validated age data would be required to assess age-related marketing preferences.

4. Which plan is most commonly selected, and by which age group?

The most commonly selected subscription plan was “Premium”, followed by “Pro” and “Basic”. This suggests that users prefer higher-tier plans, potentially due to additional features or perceived value. The age summary shows median user age of 34 (mid-30s), indicating that most customers selecting plans are within the working-age adult group. Although plan selection was not directly segmented by specific age group, the overall age distribution suggests that “Premium” plan users are mostly within this central age range. Missing age data limits more detailed age-based analysis.

5. (Optional) Which plan’s users are most likely to contact support?

Based on the support ticket analysis, users on “Basic” plan show the highest level of early support contact, in particular South and East regions. This suggests that customers on lower-tier may experience more challenges or need guidance after signing up. The “Pro” and “Premium” plans show more notable support activity, but at lower levels compared to the “Basic” plan. Overall, early support demand appears to be more focused among lower-tier subscriptions, this can be used to improve onboarding materials or in-app guidance for “Basic” plan users to reduce relying on early support.

Recommendations

Based on the analysis, marketing efforts should continue prioritising YouTube as a key acquisition source, as it consistently gives the highest number of new users. Increasing investment in video-based campaigns or refining existing content strategies may further improve user acquisition performance.

Given the popularity of higher-tier subscription plans such as “Premium” and “Pro”, the business should continue highlighting the value of advanced features during onboarding and promotional campaigns. Clear promotions of premium benefits can encourage upgrades and potentially maximise revenue.

Finally, improving data collection at the sign-up stage, particularly for region and age attribute field can improve the quality of future analysis. Making certain fields mandatory or validating inputs can reduce missing or inaccurate data leading to more reliable decision-making.

Data issues or risks

In the dataset, a key data quality issue is the presence of missing and unreliable demographic data, in particular within age and region attributes. Several records were missing region values, while the age field contained missing entries and at least one unrealistic value, which reduces the confidence in age-based analysis. This limits the accuracy of demographic insights and may impact marketing or targeting decisions. To address this issue, stricter data validation rules could be applied at the sign-up stage, such as mandatory selection of region and numeric validation for age inputs. For future analyses, improving data capture at the source would enhance data reliability and support more accurate reporting.